LAB EXPERIMENT RECORD

Set 19: Survey Responses Analysis

Department	CSE-DS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Day	Thursday

Dataset: Survey Responses

Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

1. Grouped Bar Chart

Aim

To create a grouped bar chart to visualize the distribution of answers for Question 1 and label the chart elements.

R Code

```
survey_id <- c(1, 2, 3)

question1 <- c("A", "B", "C")

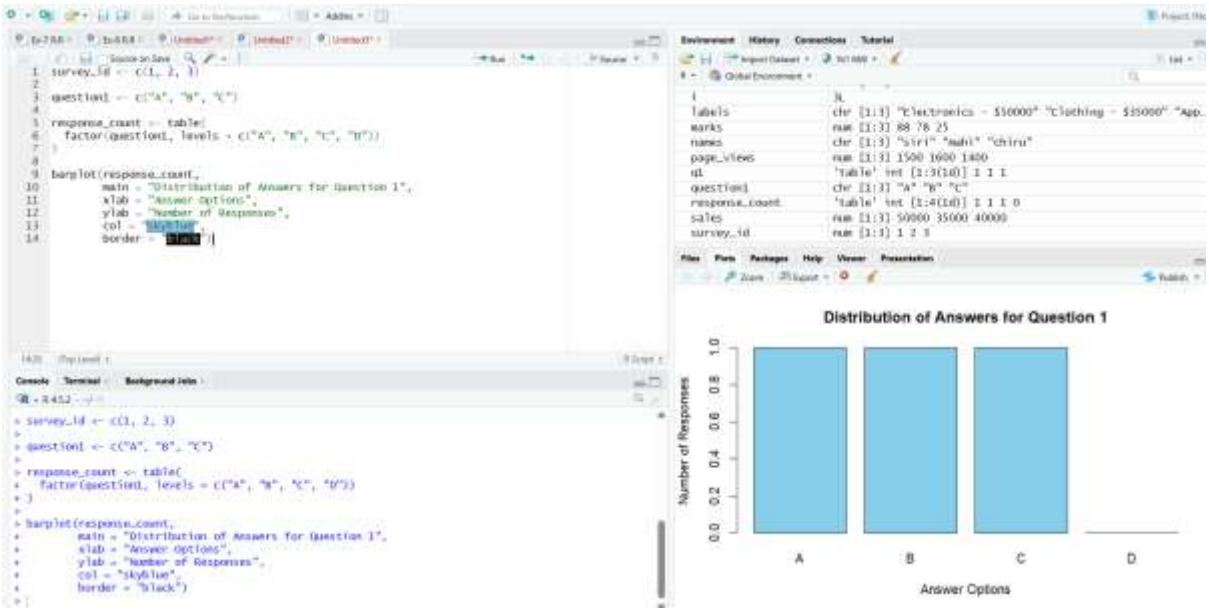
response_count <- table(

  factor(question1, levels = c("A", "B", "C", "D"))

)
```

```
barplot(response_count,
        main = "Distribution of Answers for Question 1",
        xlab = "Answer Options",
        ylab = "Number of Responses",
        col = "skyblue",
        border = "black")
```

Output



A bar chart displaying the number of responses for each answer option (A, B, C, and D) for Question 1.

Observation

Answer A, B, and C each received one response, while answer D received no responses. Therefore, all three recorded answer options have equal frequency.

2. Stacked Bar Chart

Aim

To generate a stacked bar chart representing the overall distribution of responses for all three questions.

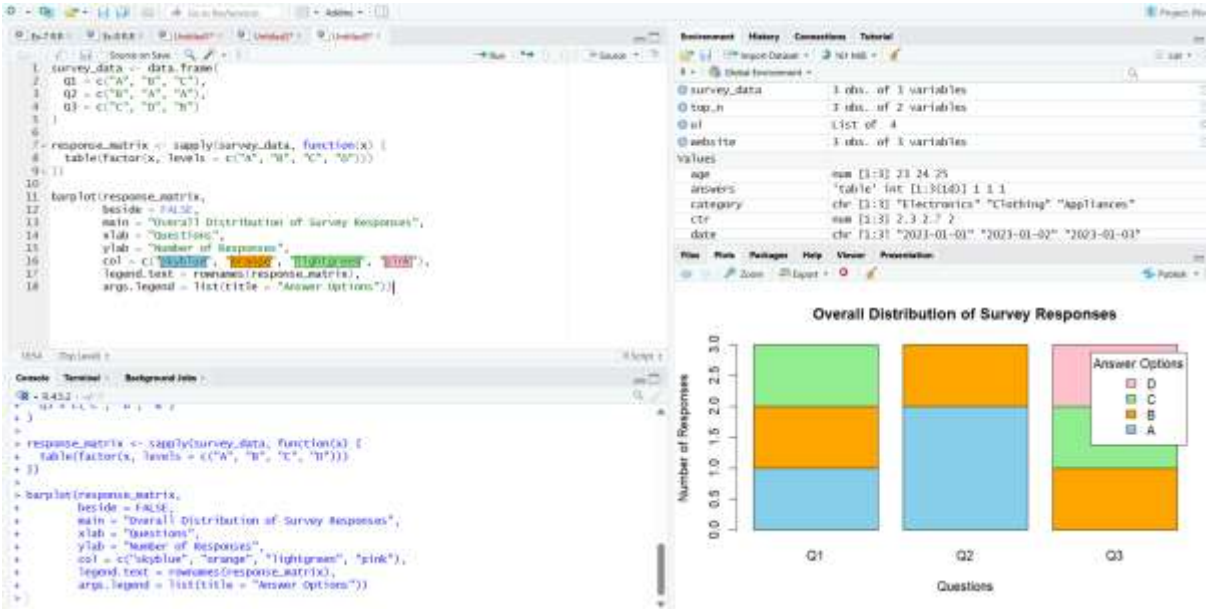
R Code

```
survey_data <- data.frame(
  Q1 = c("A", "B", "C"),
```

```
  Q2 = c("B", "A", "A"),
  Q3 = c("C", "D", "B")
)
response_matrix <- sapply(survey_data, function(x) {
  table(factor(x, levels = c("A", "B", "C", "D")))
})

barplot(response_matrix,
        beside = FALSE,
        main = "Overall Distribution of Survey Responses",
        xlab = "Questions",
        ylab = "Number of Responses",
        col = c("skyblue", "orange", "lightgreen", "pink"),
        legend.text = rownames(response_matrix),
        args.legend = list(title = "Answer Options"))
```

Output



A stacked bar chart showing the distribution of answer options A, B, C, and D across Questions 1, 2, and 3.

Observation

Question 1 contains one response each for A, B, and C. Question 2 contains two responses for A and one for B. Question 3 contains one response each for B, C, and D. Answer A is the most frequent response overall, with three responses.

3. Survey Responses Table

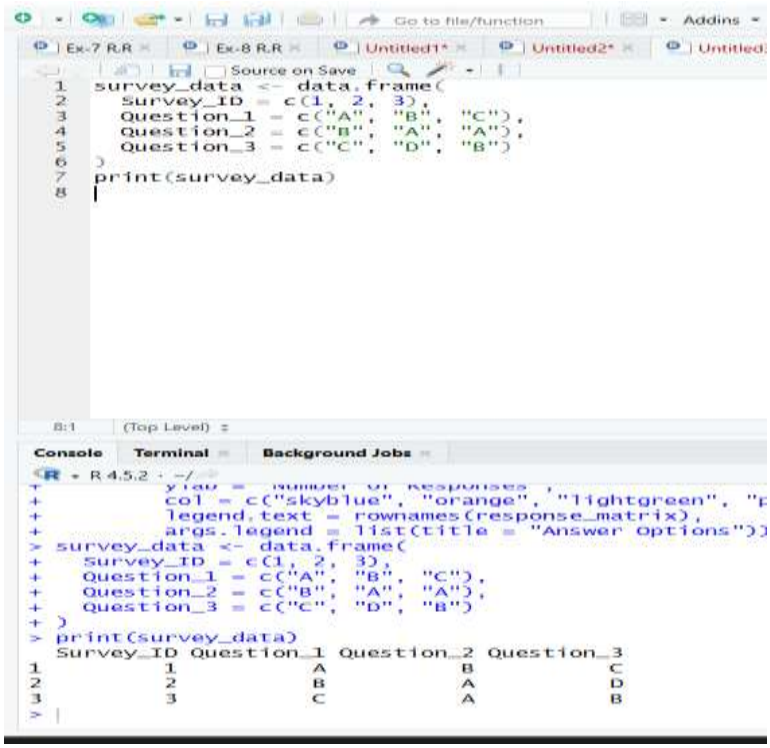
Aim

To create a table displaying the survey response data for each survey, including Survey ID and responses to all three questions.

R Code

```
survey_data <- data.frame(
  Survey_ID = c(1, 2, 3),
  Question_1 = c("A", "B", "C"),
  Question_2 = c("B", "A", "A"),
  Question_3 = c("C", "D", "B")
)
print(survey_data)
```

Output



Observation

The table clearly displays the responses submitted by all three surveys. Each row represents one survey, and each question column shows the corresponding answer.

4. Tableau Dashboard

Aim

To develop an interactive Tableau dashboard combining the grouped bar chart, stacked bar chart, and survey responses table for exploring survey data.

Tableau Procedure

1. Open Tableau.
2. Import the Survey Responses dataset.
3. Create a grouped bar chart using the Question 1 answer options and their response counts.
4. Add a suitable chart title and axis labels.
5. Create a stacked bar chart showing the response distribution for Questions 1, 2, and 3.
6. Assign different colors to answer options A, B, C, and D.
7. Create a table containing Survey ID, Question 1, Question 2, and Question 3.
8. Select **Dashboard → New Dashboard**.
9. Drag the grouped bar chart into the dashboard.
10. Drag the stacked bar chart into the dashboard.
11. Drag the survey responses table into the dashboard.
12. Arrange the charts and table neatly.
13. Add suitable titles, labels, and filters for interactive exploration.
14. Save the dashboard.

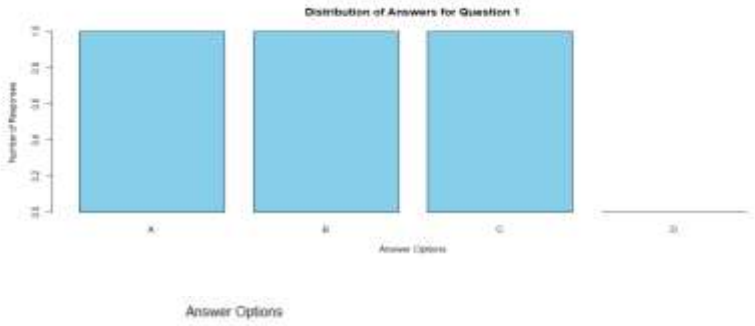
Output

Survey Responses Analysis Dashboard

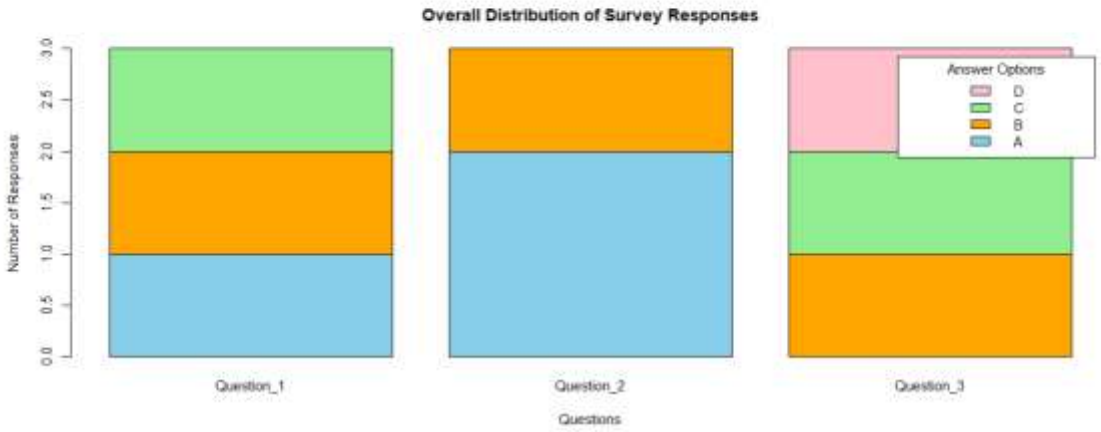
Select Question:

Question 1

Grouped Bar Chart - Question 1



Stacked Bar Chart - All Questions



Survey Responses Table

Survey_ID	Question_1	Question_2	Question_3
1.00	A	B	C
2.00	B	A	D
3.00	C	A	B

An interactive dashboard displaying the distribution of Question 1 answers, the overall response distribution for all three questions, and the survey responses table.